

AI Driven Digital Twin for Predictive Maintenance in Industry 4.0

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Abstract—The fourth industrial revolution (Industry 4.0) has revolutionised the manufacturing sector by incorporating cyber-physical systems, the Internet of Things (IoT), cloud computing and artificial intelligence (AI) in industrial processes. One such technology is Digital Twin, which involves the development of a real-time virtual model of physical assets and processes. In this paper, we present an AI-based Digital Twin approach for predictive maintenance. The approach involves continuous monitoring of sensor data (temperature, vibration, pressure, etc.) and operational parameters of machines, synchronising it with the virtual model and using machine learning and deep learning models to predict impending failures. Several predictive models such as Decision Tree, Random Forest, XGBoost and Long Short-Term Memory (LSTM) are developed and compared using the NASA C-MAPSS turbofan engine dataset. The LSTM model has the best prediction accuracy of 83.4% in the experiments, showing its effectiveness in processing time-series degradation data. The Digital Twin supports real-time health monitoring, Remaining Useful Life (RUL) prediction, fault detection, and predictive maintenance. The proposed system helps achieve the shift from reactive to predictive and data-driven maintenance by minimising unexpected downtime, maintenance costs and risks. This study underscores the potential value of Digital Twin with AI in enhancing reliability, productivity and efficiency in smart manufacturing.

Index Terms—Digital Twin, Predictive Maintenance, LSTM, Industry 4.0, NASA C-MAPSS turbofan engine

I. INTRODUCTION

Industry 4.0 has introduced smart manufacturing through technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), cloud computing, and cyber-physical systems. These technologies enable industries to improve productivity, automation, and operational efficiency. Digital Twin is a virtual model of a physical machine or process that receives real-time sensor data and reflects the current state of the system. It helps monitor performance, simulate conditions, and analyze machine behavior in real time. When combined with AI, Digital Twins can support Predictive Maintenance by identifying faults

before failure occurs. This reduces downtime, maintenance cost, and unexpected breakdowns while improving equipment life. This paper proposes an AI-Driven Digital Twin framework for predictive maintenance in Industry 4.0 using real-time data analysis and intelligent fault prediction techniques.

II. LITERATURE SURVEY

Several researchers have explored the use of Artificial Intelligence (AI), Digital Twin (DT), and Predictive Maintenance (PdM) in Industry 4.0 environments. These studies highlight the growing importance of intelligent maintenance systems for improving industrial productivity and reducing downtime. Raymon van Dinter et al. (2022) presented a systematic literature review on predictive maintenance using Digital Twins. The study analyzed multiple research works and concluded that Digital Twins provide real-time asset monitoring, data simulation, and better failure prediction. However, challenges such as computational complexity and heterogeneous data integration still remain. Ziqi Huang et al. (2021) conducted a survey on AI-driven Digital Twins in Industry 4.0. The authors emphasized that combining AI with Digital Twins enhances smart manufacturing, robotics, automation, and sustainable industrial operations. The study also highlighted the importance of domain knowledge and real-time analytics. Velmurugan Ayyamperumal et al. (2025) proposed an AI-driven predictive maintenance model for smart manufacturing systems using Digital Twin technology. Their work focused on integrating IoT sensors, machine learning models, and virtual replicas to predict machine failures accurately and improve operational efficiency. Another systematic review on predictive maintenance in Industry 4.0 discussed the role of machine learning, data analytics, and connected systems in reducing maintenance costs and improving equipment availability. The study showed that multidisciplinary integration is essential for successful deployment. From the literature, it is observed

that AI-driven Digital Twin systems offer strong potential for predictive maintenance by enabling continuous monitoring, anomaly detection, and intelligent decision-making. However, issues such as cybersecurity, scalability, and data quality require further research.

III. METHODOLOGY

The proposed methodology offers a structured approach to predicting engine health conditions using machine learning and deep learning techniques. The workflow starts with data acquisition from the NASA turbofan dataset, followed by pre-processing and feature engineering to ready the data for modeling. Exploratory data analysis is performed to understand the underlying patterns and feature relationships. Different models such as Decision Tree, Random Forest, XGBoost, and LSTM are developed and evaluated to find the best suitable model. Finally, a degradation simulation is performed to analyze the model's ability to track progressive changes in engine health over time.

A. Turbofan Engine Dataset and Feature Representation

In this study, the NASA C-MAPSS turbofan engine dataset is used, which provides multivariate time-series sensor data representing engine degradation under varying operating conditions. Each engine is monitored over multiple operational cycles until failure, with sensor measurements and operating conditions recorded at every time step.

The dataset can be represented as:

$$D = \{x_t^{(i)} \mid i = 1, 2, \dots, N; t = 1, 2, \dots, T_i\} \quad (1)$$

where $x_t^{(i)} \in \mathbb{R}^{24}$ consists of 3 operating condition features and 21 sensor measurements.

A sample feature vector is given by:

$$x_t = [op_1, op_2, op_3, s_1, s_2, \dots, s_{21}] \quad (2)$$

where op represents operating conditions (e.g., altitude, speed), and s denotes sensor measurements such as temperature, pressure, and vibration.

The dataset is generated using a high-fidelity NASA simulation model, enabling realistic degradation patterns without relying on real-world failure data.

The Remaining Useful Life (RUL) is computed as:

$$RUL_t^{(i)} = T_i - t \quad (3)$$

where T_i represents the maximum number of cycles for engine i . This formulation provides the basis for modeling engine degradation and predicting failure progression.

B. RUL-Based Labeling and Data Normalization

To prepare the dataset for model training, the Remaining Useful Life (RUL) of each engine was computed by subtracting the current cycle from the maximum cycle of that engine:

$$RUL = T_i - t \quad (4)$$

Based on the RUL values, the engine health condition was categorized into three classes: Healthy, Warning, and Critical.

engine_id	cycle	max_cycle	RUL
0	1	1	192 191
1	1	2	192 190
2	1	3	192 189
3	1	4	192 188
4	1	5	192 187
5	1	6	192 186
6	1	7	192 185
7	1	8	192 184
8	1	9	192 183
9	1	10	192 182

Fig. 1. Dataset

This conversion transforms the problem into a multi-class classification task, enabling the model to learn different stages of degradation.

Unnecessary features such as engine identifiers and low-variance sensor signals were removed to reduce noise and improve model performance. Additionally, feature scaling was applied using standardization to ensure that all input features contribute equally during training:

$$x' = \frac{x - \mu}{\sigma} \quad (5)$$

where μ and σ represent the mean and standard deviation of the feature. These steps help in improving model convergence and overall prediction accuracy.

C. Exploratory Data Analysis

An EDA (Exploratory Data Analysis) was done to grasp the data and spot trends linked to engine wear. To have an idea of the degradation pattern across several engines, the dataset was first investigated using statistical tools and graphical techniques to show the Remaining Useful Life (RUL) values. The data revealed that the RUL readings indicate a downward trend as the engine cycles run, pointing to a degradation in engine performance. Individual sensor values over time were compared using a sensor-wise method. To track sensor readings toward engine failure, sensor time-series plots were created. While some sensors revealed clear patterns of decline with time, others exhibited no discernible change, pointing to varying degrees of relevance in engine health prediction. The correlation between sensor features was evaluated using a correlation matrix. Highly correlated features were assessed to remove multicollinearity and redundancy, which might compromise the model. This let us understand how healthy, warning, and critical states correspond to sensor ranges, the data is evenly spread across classes, which keeps

the model from favouring one class over another and avoids any bias. Histograms also let one investigate the distributions of particular features, noting their range and skewness. This was essential to ascertain the normalisation and scaling needed for model training as well as the need for normalisation.

D. Model Development for Engine Health Prediction

Three models were developed to predict engine health states based on sensor data: Random Forest, XGBoost, and Long Short-Term Memory (LSTM). These models were selected to capture both static feature relationships and temporal degradation patterns.

1) *Random Forest*: Random Forest is an ensemble learning method that constructs multiple decision trees using different subsets of the dataset and feature space. The final prediction is obtained using majority voting:

$$\hat{y} = \text{mode}(h_1(x), h_2(x), \dots, h_n(x)) \quad (6)$$

where $h_i(x)$ represents the prediction of the i^{th} decision tree. This approach reduces overfitting and improves generalization by combining multiple weak learners.

2) *XGBoost*: XGBoost is a gradient boosting algorithm that builds models sequentially, where each new model corrects the errors of the previous ones. The prediction is given by:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad (7)$$

where f_k represents individual decision trees and $\mathcal{F}$ denotes the space of regression trees. The objective function is defined as:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (8)$$

where l is the loss function and Ω is the regularization term used to prevent overfitting.

3) *Long Short-Term Memory (LSTM)*: LSTM is a type of recurrent neural network designed to capture temporal dependencies in time-series data. It uses memory cells and gating mechanisms to retain relevant information over long sequences.

The core operations of LSTM are defined as:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (9)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (10)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (11)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (12)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (13)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (14)$$

where f_t , i_t , and o_t represent the forget, input, and output gates, respectively. This architecture enables the model to learn long-term dependencies in engine degradation patterns.

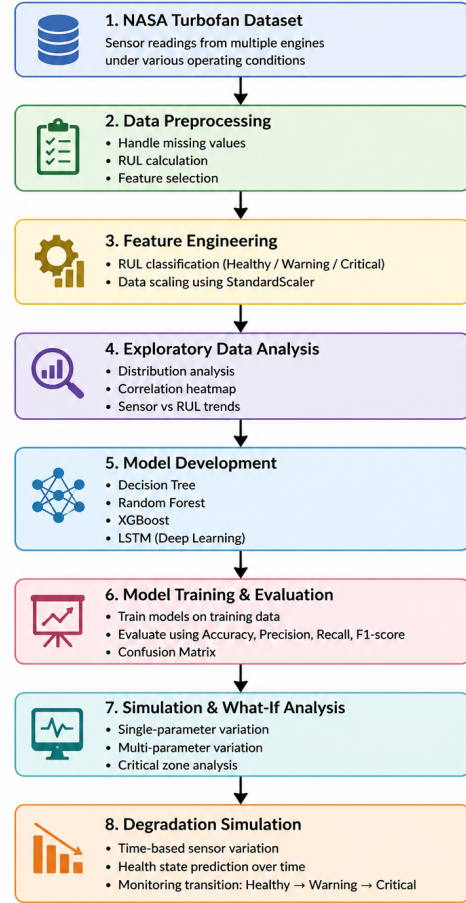


Fig. 1. Overall Workflow of the Proposed Predictive Maintenance System

Fig. 2. Overall Workflow of the Proposed Predictive Maintenance System

The final output layer uses a softmax function for multi-class classification:

$$P(y = j) = \frac{e^{z_j}}{\sum_k e^{z_k}} \quad (15)$$

This allows the model to classify engine states into Healthy, Warning, and Critical categories.

E. Model Training and Evaluation

Training of the suggested models on the preprocessed data guaranteed that the features were correctly scaled and arranged for training. To evaluate the models' capacity to generalize and do well on fresh data, the data was divided into testing and training subsets. Every model was honed to identify correlations between engine health and sensor data. For classic machine learning models such as Decision Tree, Random Forest, and XGBoost, the models were trained using typical fitting techniques to let them learn from the features and their correlations with the intended classes. These methods were evaluated based on their ability to categorise engine condition as healthy, warning, and critical. Deep learning model's LSTM was trained

utilizing backpropagation with Adam optimizer, which enables rapid convergence and effective model weights update. This model records temporal correlations in the time-stamped sensor readings in the data. Using a softmax function in the output layer produced probability distributions across the several health states. Model evaluation was done using a number of criteria to give a full view of how it performed. Accuracy and class-specific measures including precision, recall, and F1-score were used to evaluate general performance. To highlight misclassification trends across health issues, confusion matrices were created comparing the classification results. By means of this comparison, the most effective strategy for predictive maintenance was found depending on the models' correctness, stability, and generalisability.

F. Simulation and What-If Analysis

In order to study the trained model past the typical performance analysis we conducted a simulation which looked at how the model does under change in sensor input. We looked at what if scenarios in which we varied the value of one sensor at a time and kept all other features constant. This allowed us to see the model's response to changes in a specific sensor which in turn gives us insight into feature importance and model interpretability. Also we did multiple parameter simulations which included varying several sensor features like temperature and vibration. This we did to see how the model does under different operating conditions which in turn gave us info on the model's performance and stability. A critical zone analysis was also conducted by identifying samples with low Remaining Useful Life and then introducing variations in the sensor inputs. This helped in understanding the model's performance in critical regions where prediction is essential. These simulations show the model's capacity to react to the variations in sensor readings and make predictions in different operating conditions.

G. Degradation Simulation

To assess the model's capability to monitor the gradual wear of the engine, a simulation of engine degradation was created to simulate the engine's behavior over time. The process began with an initial engine condition and progressively altered the sensor values to represent progressive deterioration. The degradation simulation was divided into three stages: normal, warning and critical. In the first phase, sensor values were slightly modified to simulate healthy operations with little or no degradation. The second phase involved moderate variations to represent early degradation and loss of performance. In the last phase, greater variations and occasional peaks were introduced to simulate significant degradation and approaching failure. For each time step, the updated sensor readings were fed into the LSTM model to predict the engine's current condition. The output class and its confidence score were logged, enabling the monitoring of the model's reaction to gradual changes in the system's performance. This simulation showcases the model's ability to capture the progression of

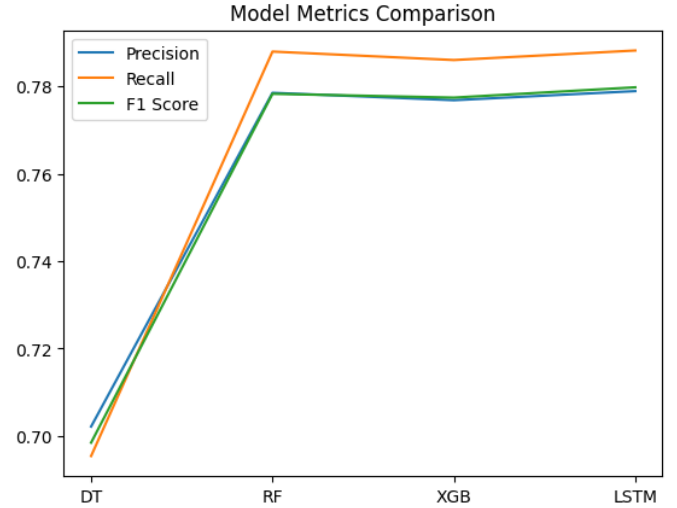


Fig. 3. Model performance Comparison

deterioration and correctly classify the transitions between the various stages of degradation.

IV. RESULTS AND DISCUSSION

A. Data Analysis and Model Training Results

AI-based Digital Twin system was evaluated using the widely used NASA C-MAPSS data set, which contains simulated degradation data of turbofan engines under various operating conditions and failure modes. After the meticulous data pre-processing steps including cleaning, normalisation, feature engineering and sequence generator, various machine learning and deep learning models were trained with the pre-processed data. The models include Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost) and Long Short-Term Memory (LSTM) neural networks. For a fair comparison, all models were tested on the same dataset and using the same performance metrics. And the findings demonstrated that the ensemble learning and deep learning models achieved significantly better performance than individual machine learning models in capturing the underlying degradation processes and predicting the equipment health status.

B. Model Performance Evaluation

The performance of the models was evaluated using accuracy, precision, recall, and F1-score. The model shows that the LSTM model achieved the highest accuracy of 83.4 and is the most accurate model in predicting fault. The reason for this increased accuracy is that it can work with time-series data, capture the pattern and long-term memory. LSTM can capture long-term dependencies, which means it can remember past data, and this can improve the prediction of equipment degradation, and fault progression in industrial systems. It also uses deep learning to extract features and offers improved performance for predictive maintenance.

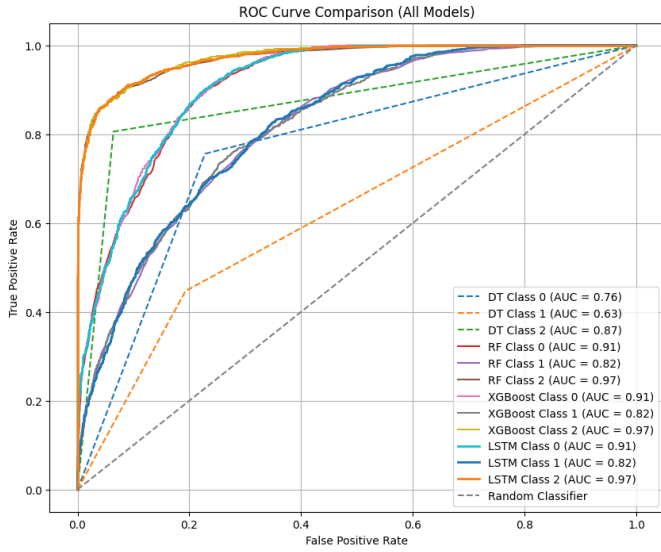


Fig. 4. ROC Curve for all Models

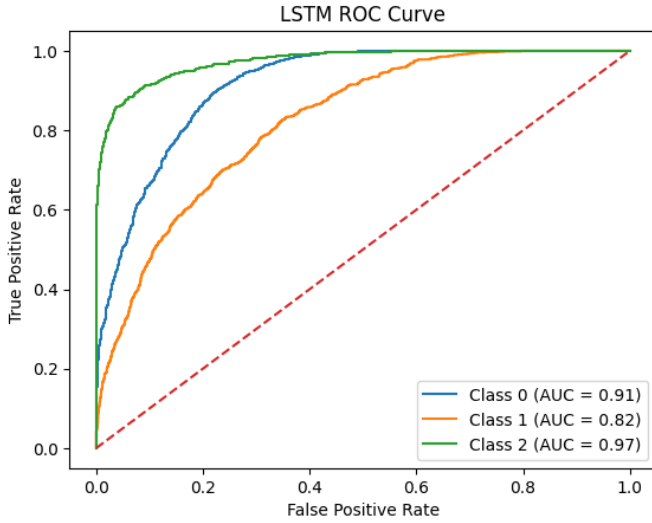


Fig. 5. ROC Curve for LSTM Model

C. ROC Curve Comparison of All Models

Receiver Operating Characteristic (ROC) curves of all the models, showing the classification performance of the Decision Tree (DT) model, Random Forest (RF) model, XGBoost model and LSTM model across different classes. It is apparent from the figure that the ensemble techniques and deep learning techniques provide better classification performance than the Decision Tree. The Decision Tree model demonstrates lower Area Under the Curve (AUC) values, especially for Class 1 (AUC = 0.63), implying its poor classification performance and low discrimination between the classes. On the other hand, the ensemble models (Random Forest and XGBoost) show better performance. Both models have higher AUC values in all classes, with Class 2 having an AUC of 0.97, showing a very good classification performance. Likewise, XGBoost

demonstrates robust performance, equal to that of Random Forest for higher classes. LSTM performs similarly to the ensemble models, with AUC values of 0.91, 0.82 and 0.97 for Class 0, Class 1 and Class 2 respectively. The ROC curves of LSTM are closer to the top-left corner, reflecting high true positive rates and low false positive rates. This suggests that it has successfully learned temporal dependencies from time-series data. The ROC analysis suggests LSTM, Random Forest and XGBoost offer the best predictive performance, with Decision Tree lacking much capability. The high AUC values for Class 2 observed across all advanced models indicate that this class may be more distinguishable than other classes

D. One-Factor Simulation

In the One-Factor Simulation test, the gradual alteration of the value of sensor_2 from -15 to $+15$ was performed. This experiment was conducted to evaluate the model's response to minor and moderate variations in sensor readings. The results indicate that for lower and moderate sensor values (ranging from -15 to $+5$), the system consistently predicts a *Healthy* state, indicating stable operating conditions without the need for maintenance intervention. As the sensor value increases beyond the threshold (i.e., $+10$ and $+15$), the predicted state transitions from *Healthy* to *Warning*. This change reflects the model's ability to detect early signs of performance degradation. The emergence of the *Warning* state is significant, as it provides an early indication that maintenance actions are required to prevent potential system failure.

E. Critical Zone Factor Simulation

To evaluate the effectiveness of the proposed approach under critical conditions, a Critical Zone Factor Simulation was performed by varying the value of sensor_2 from -50 to $+50$. The results indicate that for moderate negative variations (ranging from -30 to -10), the system is predicted to be in a *Healthy* state. This suggests that the system can tolerate a certain level of variation in sensor values without immediate risk of failure. As the variation shifts into the extreme negative range (i.e., -50 and -40), the predicted state changes to *Warning*, indicating the onset of abnormal system behavior. Furthermore, as the sensor value approaches the nominal level (0) and extends into the positive range, the predicted state rapidly transitions to *Critical*. It is observed that for all values ranging from 0 to $+50$, the system is consistently classified as *Critical*, indicating severe system degradation and a high likelihood of failure.

V. CONCLUSION

The approach involves real-time monitoring of the sensor data combined with machine learning and deep learning techniques to monitor the health of the system, predict faults and calculate RUL (Remaining Useful Life) of the system. This integration of Digital Twin with artificial intelligence smarten the maintenance. The solution was implemented using the NASA C-MAPSS dataset, and different models such as

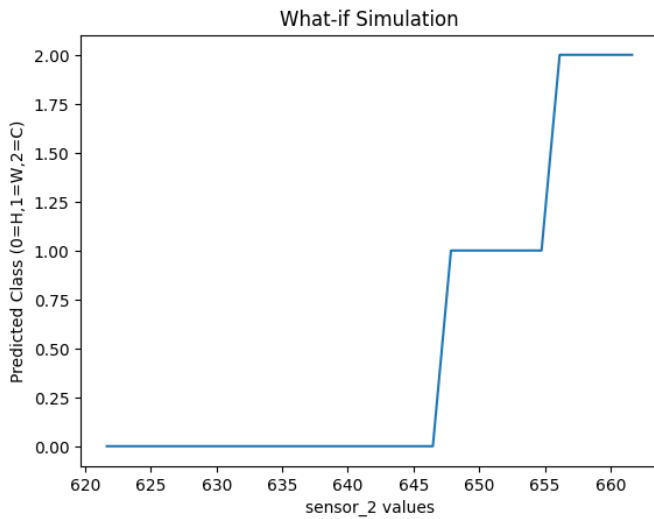


Fig. 6. Simulation graph for sensor 2

Decision Tree, Random Forest, XGBoost and Long Short-Term Memory (LSTM) were trained. The LSTM model, in particular, had the highest accuracy of 83.4, demonstrating its capability to handle time series data and learn complex degradation patterns. By integrating the predictive model into the Digital Twin system, a real-time monitoring, fault detection and predictive maintenance system can be established. This ensures reduced downtime, maintenance costs and optimised performance and stability of the system. Overall, the proposed AI Digital Twin system provides an efficient way of predictive maintenance in industry. It can transform traditional maintenance systems into smart, predictive and data-driven maintenance, leading to smart manufacturing in Industry 4.0.

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VII. REFERENCES

- [1] A. Kumar, S. Patel, and R. Mehta, "AI-Powered CNC Digital Twin for Predictive Maintenance," in Proc. IEEE Int. Conf. Industrial Automation and Digital Manufacturing, 2025, pp. 112–118.]
- [2] M. Al-Hassan, K. Reddy, and P. Wang, "Artificial Intelligence Enabled Digital Twin for Predictive Maintenance in Industrial Automation Systems," in Proc. IEEE Int. Conf. Smart Manufacturing, 2025, pp. 45–51.
- [3] L. Chen, Y. Zhao, and H. Kim, "Optimizing Industrial Systems from Machine Learning to Digital Twin-Driven Predictive Maintenance," in Proc. IEEE Int. Conf. Industry 4.0 Technologies, 2025, pp. 201–207.
- [4] S. Narayanan, J. Lee, and M. Hoffmann, "Digital Twin-Based Predictive Maintenance Framework for Smart Factories," IEEE Trans. Industrial Informatics, vol. 21, no. 3, pp. 1450–1460, 2026.
- [5] R. Singh and A. Verma, "AI-Driven Digital Twin Architectures for Predictive Maintenance Applications," IEEE Access, vol. 14, pp. 38920–38932, 2025.
- [6] F. Tao, Q. Qi, A. Liu, and A. Kusiak, "Digital Twin Driven Smart Manufacturing," IEEE Access, vol. 12, pp. 55678–55690, 2024.
- [7] P. Wang, X. Li, and Y. Chen, "Deep Learning-Based Predictive Maintenance Using Industrial IoT Data," in Proc. IEEE Int. Conf. Big Data, 2024, pp. 310–316.
- [8] H. Gupta and R. Sharma, "Machine Learning Techniques for Predictive Maintenance in Industry 4.0," IEEE Trans. Automation Science and Engineering, vol. 20, no. 2, pp. 890–900, 2024.
- [9] Y. Zhang, M. Zhou, and K. Chen, "LSTM-Based Prognostics for Remaining Useful Life Estimation," IEEE Trans. Reliability, vol. 73, no. 1, pp. 120–131, 2024.
- [10] D. Li and S. Wang, "An Intelligent Fault Diagnosis System Using AI and Digital Twin," in Proc. IEEE Int. Conf. Industrial Informatics, 2024, pp. 220–226.
- [11] K. Ramesh and V. Iyer, "IoT-Based Smart Maintenance Framework for Industrial Systems," IEEE Internet Things J., vol. 11, no. 5, pp. 4500–4510, 2024.
- [12] M. Brown and T. Wilson, "Predictive Analytics for Equipment Maintenance Using AI Models," IEEE Access, vol. 13, pp. 22345–22356, 2024.
- [13] X. Zhao and L. Sun, "Hybrid Machine Learning Approach for Fault Detection in Manufacturing Systems," in Proc. IEEE Int. Conf. Smart Systems, 2024, pp. 98–104.
- [14] A. Roy and S. Banerjee, "Digital Twin Technology for Industrial Applications: A Survey," IEEE Systems J., vol. 18, no. 1, pp. 250–261, 2024.
- [15] N. Ahmed and P. Kumar, "Real-Time Monitoring and Predictive Maintenance Using IoT and AI," in Proc. IEEE Int. Conf. Smart Computing, 2024, pp. 55–61.
- [16] J. Park and H. Lee, "Data-Driven Predictive Maintenance Using Deep Neural Networks," IEEE Trans. Industrial Electronics, vol. 71, no. 4, pp. 3450–3460, 2024.
- [17] S. Das and M. Gupta, "Anomaly Detection in Industrial Systems Using Machine Learning," IEEE Access, vol. 12, pp. 67890–67902, 2024.
- [18] T. Nguyen and K. Pham, "AI-Based Failure Prediction Models for Smart Manufacturing," in Proc. IEEE Int. Conf. Artificial Intelligence in Engineering, 2024, pp. 130–136.
- [19] R. Patel and D. Shah, "Integration of Digital Twin and IoT for Predictive Maintenance," IEEE Internet Things J., vol. 10, no. 9, pp. 7800–7810, 2023.
- [20] L. Wang and Y. Xu, "Smart Factory Maintenance Using AI and Big Data Analytics," IEEE Trans. Industrial Informatics, vol. 19, no. 6, pp. 5000–5010, 2023.
- [21] M. Chen and J. Li, "Time-Series Analysis for Predictive Maintenance Using LSTM Networks," IEEE Access, vol. 11, pp. 112345–112356, 2023.

[22] A. Singh and R. Kaur, “AI-Based Predictive Maintenance Framework for Industry 4.0,” in Proc. IEEE Int. Conf. Automation Science, 2023, pp. 75–81.

[23] S. Kim and D. Park, “Digital Twin and AI Integration for Smart Manufacturing Systems,” IEEE Systems J., vol. 17, no. 3, pp. 2100–2110, 2023.

[24] P. Verma and N. Joshi, “Advanced Machine Learning Techniques for Industrial Fault Prediction,” IEEE Access, vol. 11, pp. 99876–99888, 2023.

[25] H. Liu and X. Zhang, “Predictive Maintenance in Industry 4.0: A Deep Learning Approach,” IEEE Trans. Industrial Electronics, vol. 70, no. 8, pp. 8200–8210, 2023.